

SMART BANK STATEMENT ANALYSER — MASTER DEVELOPMENT PROMPT

Build a professional **Smart Bank Statement Analyser** for accountants, auditors, finance professionals and businesses.

1. INPUT OPTIONS

Provide three upload options:

- Excel/CSV bank statement
- PDF bank statement
- Bank-statement image/scanned document

Allow multiple files and support common Indian bank statement formats.

2. DATA EXTRACTION

Automatically identify and extract:

- Transaction Date
- Value Date
- Description/Narration
- Cheque/Reference/UTR Number
- Debit
- Credit
- Balance
- Account/Bank details where available

For scanned PDFs/images, use OCR. Preserve the original transaction description without unnecessary alteration.

3. INTELLIGENT PROCESSING

Automatically:

- Detect opening and closing balance.
- Identify credits and debits.
- Standardise dates and amounts.
- Remove duplicate transactions.
- Detect missing/invalid rows.
- Reconcile running balance wherever possible.
- Flag extraction or reconciliation errors for user review.

Never invent missing transaction data. Mark uncertain fields as **"Review Required."**

4. TRANSACTION CLASSIFICATION

Use narration and transaction patterns to intelligently classify transactions into categories such as:

- Salary
- Sales/Business Receipt
- Purchase/Business Expense
- Bank Charges
- Interest
- Loan/EMI
- Cash Deposit
- Cash Withdrawal
- UPI
- NEFT
- RTGS
- IMPS
- Cheque
- Transfer
- Investment
- Tax/GST/Income Tax
- Refund
- Other

Allow the user to edit categories.

5. HIGH-VALUE TRANSACTION DETECTION

Automatically identify high-value transactions using:

- User-defined threshold, OR
- Intelligent threshold based on statement transaction patterns.

Display high-value transactions prominently and **highlight the complete transaction row in RED** in the Excel output.

Also identify:

- Unusually large transactions
- Large cash transactions
- Repeated large transactions
- Unusual credits/debits
- Round-value transactions
- Possible related/repetitive transactions
- Transactions requiring audit review

Do NOT label a transaction as fraud merely because it is unusual. Use terminology such as **“High-Value,” “Unusual,” or “Review Required.”**

6. SMART ANALYTICS DASHBOARD

Generate a summary containing:

- Total Credits
- Total Debits
- Net Cash Flow
- Opening Balance
- Closing Balance
- Average Transaction Value
- Highest Credit
- Highest Debit
- Number of Transactions
- High-Value Transaction Count
- Cash Deposit/Withdrawal totals
- UPI/NEFT/RTGS/IMPS totals where identifiable

Include suitable charts for monthly inflow/outflow, transaction categories and high-value transactions.

7. SMART EXCEL EXPORT

Create a professional Excel workbook containing:

Sheet 1 — Executive Summary KPI dashboard and charts.

Sheet 2 — All Transactions Structured table with filters, sorting, frozen headers and proper number/date formatting.

Sheet 3 — High-Value Transactions Only high-value/unusual transactions with reason for flagging.

Sheet 4 — Category Analysis Category-wise transaction count and amount.

Sheet 5 — Monthly Analysis Month-wise credit, debit and net movement.

Use **RED highlighting for high-value transactions**, with a legend explaining the highlighting rules.

8. PDF ANALYSIS REPORT

Generate a professional PDF report containing: 1. Report title and statement period 2. Executive summary 3. Account/statement information 4. Credit/debit analysis 5. Monthly cash-flow analysis 6. Category analysis 7. High-value transaction table 8. Unusual transaction observations 9. Data-quality/reconciliation exceptions 10. Audit review points 11. Disclaimer

Use clear tables, charts and professional formatting.

9. USER CONTROLS

Provide controls for:

- High-value threshold

- Date range
- Transaction category
- Minimum amount filter
- Credit/Debit filter
- Export Excel
- Export PDF
- Reset analysis

10. IMPORTANT ACCURACY RULES

Maintain complete transaction-level traceability to the uploaded statement.

Never fabricate, estimate or silently modify transaction amounts, dates, descriptions or balances.

Where OCR/extraction confidence is low, clearly mark the transaction **“Review Required.”**

Clearly distinguish **fact extracted from the statement** from **AI-generated analytical observations**.

11. UI DESIGN

Create a modern, simple and professional dashboard with: **Upload → Preview → Analyse → Review → Export**

Use professional cards, tables, filters, charts and status indicators.

The application should be suitable for practical use by **accountants, auditors, finance teams and businesses** and should handle large bank statements efficiently.